MILITARY SPECIFICATION

LITHIUM TETRAFLUOROBORATE (FOR USE IN AMMUNITION)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one grade of lithium tetrafluoroborate ($LiBF_4$) for use in lithium reserve type batteries. (See 6.3).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipping and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-E11	- Wire-Cloth Sieves For Testing Purpose
ASTM-E70	- pH of Aqueous Solutions with the Glass Electrode
ASTM-E168	- Infra-red Quantitative Analysis, Rec Practices
ASTM-E203	- Water Using Karl Fischer Reagent
ASTM-E258	- Total Nitrogen in Organic Materials by Modified Kjeldahl Method
ASTM-E300	- Sampling Industrial Chemicals

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

3. REQUIREMENTS

3.1 Material. The material shall be lithium tetrafluoroborate ($LiBF_4$) conforming to the following requirements: (see 6.3)

3.1.1 Form. The material shall be in the form of white crystalline powder and shall be free of dirt, chips, grease and other foreign matter when examined as specified in 4.4.3.1.

3.1.2 X-ray diffraction. The material shall have an X-ray diffraction pattern which conforms to Table I when tested as specified in 4.4.3.2:

Table I

Peak, d-value	Range
3.35	3.30-3.40
4.25	4.17-4.33
3.97	3.88-4.05
3.16	3.10-3.22

3.1.3 Infrared Spectroscopy. The material shall conform with the infrared spectrum for $LiBF_4$ (see 6.6) and to Table II when tested as specified in 4.4.3.3.

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Table II

Peak, Microns	Range
2.94	2.85-3.05
6.15 (this peak may appear as a doublet)	6.05-6.30
9.66	9.00-10.00
16.10	15.40-16.80

3.1.4 Impurities. The impurities present in the material shall not exceed the values given in Table III, when tested in accordance with the applicable paragraph. In addition, the total of all impurities shall not exceed 2500 ppm.

Table III

Impurities	Maximum Allowable Concentration ppm	Test Paragraph
Metallic Impurities		4.4.3.4
Copper	50	
Magnesium	50	
Sodium	300	
Silver	50	
Iron	200	
Aluminum	50	
Nickel	50	
Chromium	50	
Calcium	100	
Silicon	100	
Nitrogen	50	4.4.3.5
Halogens		4.4.3.6
Chlorine	50	
Bromine	50	
Iodine	50	
Sulfur	100	4.4.3.7
Carbon	200	4.4.3.8
Hydrogen	1000	4.4.3.8
Oxygen Containing Compounds	None Detectable	4.4.3.9
Fluorides	> 50	4.4.3.10
Water	500	4.4.3.11

3.1.5 Properties. The material shall conform with the following properties when tested in accordance with the applicable test paragraph.

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Property	Requirement	Test Paragraph
pH		
in water	≥ 2.0 to 7.0	4.4.3.12
in Methyl Formate (MF)	≥ 2.5 to 7.0	
Solubility in MF,%	98	4.4.3.13
Hydrolysis/Dehydration		4.4.3.14
Methanol, % max @200hrs	12	
Dimethyl ether, % max @200hrs	0.1	
Particle Size		4.4.3.15
Thru a SS No.25, %,min	90	
Thru a SS No.120, %,min	75	

3.2 First article testing. This specification makes provisions for first article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract. (see 6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.3)
b. Quality Conformance Inspection (see 4.4)

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article shall consist of 2 pounds of $LiBF_4$ obtained by sampling as described in 4.4.2.1. The sample shall be obtained from a production lot which has been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All materials, shall be obtained from the same sources of supply as will be used in regular production.

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4.3.2 Inspections to be performed. The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.4.3 of this specification.

4.3.3 Rejection. See MIL-A-48078 (PA).

4.4 Quality Conformance Inspection

4.4.1 Lot formation. A lot shall consist of one or more batches of $LiBF_4$ produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of $LiBF_4$ that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous. The product shall be submitted for inspection in accordance with MIL-STD-105.

4.4.2 Testing. Sampling and testing shall be performed in a dry atmosphere.

4.4.2.1 Sampling. Approximately 1 pound of $LiBF_4$ shall be withdrawn from each batch to be sampled using ASTM Procedure E300-70 for solids. Samples shall be selected for inspection in accordance with MIL-STD-1235, CSP-1 Plan, Inspection Level II, AQL 6.5%. If any sample fails to meet any test requirement the batch represented by the sample shall be rejected. All batches produced between the time that the last batch was tested and accepted and the batch which failed should be tested in accordance with the applicable methods given in paragraph 4.4.3. If any of these batches fail to meet any of the test requirements, that batch shall also, be rejected. In addition, after any failure of a batch the contractor will return to 100% inspection until "i" successive batches are accepted as required by MIL-STD-1235. The samples shall be subjected to the tests listed below: (Sampling shall be performed in a dry atmosphere).

Inspection/Requirement	Defect Classification
Form & Color (see 3.1.1)	Major B
X-ray Diffraction (see 3.1.2)	Major B
Infra-red Spectroscopy (see 3.1.3)	Major B
Oxygen Containing Compounds (see 3.1.4)	Major B
Moisture (see 3.1.4)	Major B
pH (see 3.1.5)	Major B
Solubility (see 3.1.5)	Major B

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4.4.2.2 Inspection Equipment. The government reserves the right to inspect the contractor's equipment and determine that he has available and utilizes correctly, measuring and test equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. Commercial inspection equipment, shall be employed where applicable for all tests and examinations specified in 4.4.3. The contractor is responsible for assuring proper calibration procedures are followed. Government approval of inspection equipment is required prior to its use for acceptance purpose.

4.4.3 Test methods and procedures. (see 6.2) The tests in 4.4.3.4 thru 4.4.3.15 shall be performed using prescribed analytical procedures for replicate determinations given in standard analytical textbooks. Testing shall be performed in a dry atmosphere.

4.4.3.1 Form. Spread approximately 1/2 pound of the sample on a flat surface and examine the material visually for the presence of contamination and form at a magnification of 25X.

4.4.3.2 X-Ray Diffraction. An X-ray diffraction pattern of the material shall be obtained using a powder X-ray diffraction technique described in a standard for reference textbook on X-ray diffraction.

4.4.3.3 Infra-red Spectroscopy. Perform an infra-red spectrographic analysis of the material in accordance with method given in ASTM-E168.

4.4.3.4 Metallic Impurities. Determine the metallic impurities using a Atomic Absorption method described in a standard or reference textbook.

4.4.3.5 Nitrogen. Determine the nitrogen content using the standard method described in ASTM-E258-07 except that copper sulphate shall be used as the catalyst.

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4.4.3.6 Halogens. Determine the halogens present in the compound using a spectrophotometric method described in the literature or reference textbook (see 6.4).

4.4.3.7 Sulfur. Determine the sulfur present in the compound using a method appearing in the literature or a reference textbook (see 6.5).

4.4.3.8 Carbon. Determine the carbon present in the compound using a direct combustion method appearing in a reference textbook. It is suggested that the direct combustion method utilize purified nitrogen gas and a two part combination tube, 500-700mg of sample in the rear section and Copper Oxide (CuO) catalyst in the forward section and the following procedure be used.

a. Heat catalyst to 500°C.
b. Heat samples to 180°C for 1/2 hour.
c. Absorb CO_2 in an Ascarite filled cartridge and absorb water in an Anhydrone filled cartridge.
d. Weigh cartridges
e. Calculate % carbon and % hydrogen according to the following equations:

$$\%C = \frac{(\text{Wt } CO_2)\ (12)}{(4.4)\ (\text{grams of sample})} \text{ X100}$$

$$\%H = \frac{(\text{Wt } H_2O)\ (2)}{(18)\ (\text{grams of sample})} \text{ X100}$$

4.4.3.9 Oxygen Containing compounds. Determine the presence of oxygen containing compounds present in the material by using a pyrolysis mass spectrographic method which appears in a standard or a reference textbook.

4.4.3.10 Fluorides. Use a Specific Ion Electrode method which is described in a standard or reference textbook.

4.4.3.11 Water. Determine the moisture content of the material using the Karl Fischer method given in ASTM-E203.

4.4.2.12 pH. Prepare a 1 Molar Solution of the $LiBF_4$ by dissolving 93.75 g of the sample in a liter of distilled water and determine the pH of the resultant solution in accordance with the method given in ASTM-E70.

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In addition, prepare a 2 Molar solution of the $LiBF_4$ by dissolving 187.5 g of sample in a liter of Methyl Formate. Quantitatively transfer 100 ml of this solution to a 1000 ml volumetric flask and dilute to the mark with distilled water. Determine the pH of this solution in accordance with the method given in ASTM-E70.

4.4.3.13 Solubility in Methyl Formate. Transfer 18.75 grams of the sample to a tall form 200 ml beaker and add 100 ml Methyl Formate. Add a magnetic stirring bar to the beaker and allow to stir for 30 minutes, at ambient temperature, on a magnetic stirrer. During mixing seal the beaker to prevent loss of methyl formate. Remove the magnetic stirring bar and quantitatively filter the contents of the beaker thru a clean, tared glass medium porosity filtering crucible. Use a small amount of MF to aid in the transfer. Dry the crucible and contents in an oven maintained at 85°C $\pm$ 5°C for 1 hour and weigh. Calculate the percent insolubles by the following equation.

$$\% \text{ Insolubles} = \frac{\text{Wt of residue X100}}{18.75}$$

4.4.3.14 Hydrolysis/Dehydration. Determine the hydrolysis/dehydration of the 2 Molar solution prepared in 4.4.3.12 as follows:

4.4.3.14.1 Equipment

a. Injection port packed with glass wool
b. Column - Chromosorb 101, 4' x 1/8" diameter
c. Detector - Flame ionization
d. Carrier Gas - Helium at 20 cc/min

4.4.3.14.2 Procedure. Add 10% distilled water to 4 ml of the MF solution. Store this solution at 57 $\pm$ 1°C for 200 hours in a suitable glass container (5cc vial) capable of being sealed air tight with stopper and withstanding 100 psi. Remove stoppered container and measure by standard gas chromatographic techniques:

Methanol - 200 hours
Dimethyl Ether - 200 hours

4.4.3.15 Particle Size. Superimpose the sieves of the meshes required in 3.1.5 as defined in ASTM-E11-61 and attach the receiving pan to the bottom. Place 100 grams of the sample in the top sieve, mount the assembly on a shaker and operate the shaker for 5 minutes. At the end of 5 minutes weigh the granules retained in each sieve and in the pan and calculate the percentages of each.

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5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level A and Level C. Unless otherwise specified, the $LiBF_4$ shall be packed in accordance with manufacturer's commercial practice to assure acceptance by common carrier for safe delivery at first destination for immediate use.

5.2 Marking. Unless otherwise specified, the shipping containers shall be marked in accordance with MIL-STD-129, and shall include the lot number.

6. NOTES

6.1 Ordering data. See MIL-A-48078 (PA).

6.2 Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted thru the Contracting Officer to: Commander ATTN: SARPA—QA-A-P. Picatinny Arsenal, Dover, New Jersey 07801. This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.3 Material meeting the requirements of this specification was previously obtained from:

Foote Mineral Company
Route 100
Exton, Penna. 19341

6.4 Method described in F.W. Chapman Jr and R.M. Sherwood, Anal, Chem. 29 (2), 172 (1957) has been found applicable.

6.5 Methods described in J.S. Fritz, S.S. Yamamura, Anal. Chem. 29 (1) 1581 or Standard Methods of Chem. Analysis II B, 2483 have been found applicable.

6.6 A qualified infrared spectrum of $LiBF_4$ can be found in Sadtler collection of inorganic spectra under Infra-Red Spectrogram No. Y-1025K.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 6810-AA66

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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